Total marks — 95

Attempt ALL questions

MARKS | DO NOT WRITE IN THIS MARGIN

1. Elements and compounds can exist as diatomic molecules.

(a) The seven elements that exist as diatomic molecules are shown in the periodic table below.

1 H																	2 He
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og

(i) Explain why diatomic elements form non-polar molecules. **1**

(ii) Nitrogen, oxygen and fluorine are found in the second period of the periodic table.

Explain the decrease in covalent radius going from nitrogen to fluorine. **1**

(b) First ionisation energies decrease going down a group.

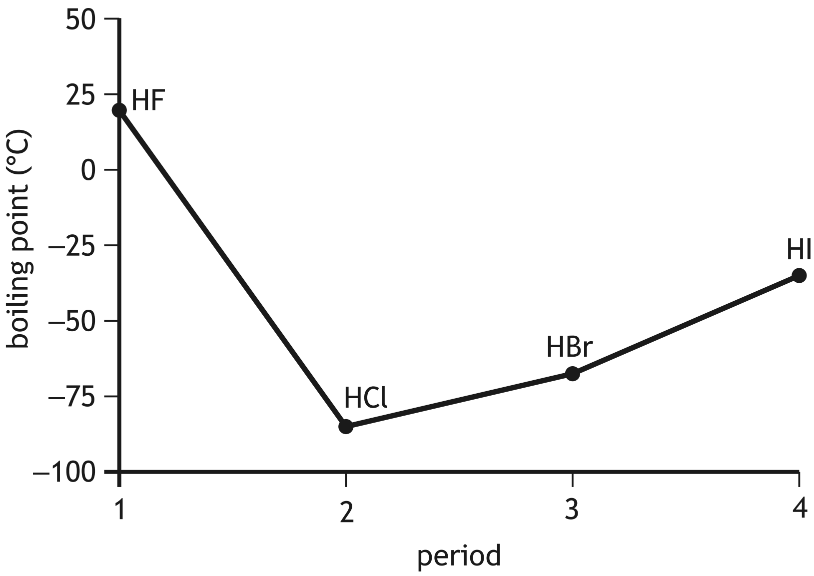
(i) State what is meant by the term *first ionisation energy*. **1**

1. (b) (continued)

(ii) Explain why the first ionisation energy of the group 7 elements decreases going down the group. **1**

(c) Hydrogen halides are diatomic molecules formed between hydrogen and the elements fluorine, chlorine, bromine and iodine.

The boiling points of the hydrogen halides are shown on the graph below.



(i) Hydrogen fluoride, HF, has the highest boiling point of the hydrogen halides.

State the name of the strongest type of intermolecular force found between hydrogen fluoride molecules and explain how this type of intermolecular force arises. **2**

[Turn over

1. (c) (continued)

(ii) The table shows the boiling points of hydrogen chloride, hydrogen bromide and hydrogen iodide.

Hydrogen halide	Boiling point (°C)
Hydrogen chloride	–85
Hydrogen bromide	–66
Hydrogen iodide	–35

Explain **fully** why the boiling point increases from hydrogen chloride to hydrogen iodide. **2**